

Machine Learning Model Evaluation

One-Page Cheat Sheet — Simple Edition

from ML_Model_Evaluation_Simple.ipynb · full course: ML_Model_Evaluation.ipynb

1 · The Confusion Matrix

Predicted: NO		Predicted: YES	
Actually: YES	TN	FP	TP
			said YES and was right — hit
Actually: NO			
			said NO and was right
			said YES but wrong — false alarm
			said NO but wrong — miss

2 · The Decision Framework

Business Problem → Cost of Errors → Important Error Type → Appropriate Metric

Model Evaluation → Threshold Selection → Business Decision

3 · Metrics at a Glance

Metric	Formula	Measures	Business question
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Overall correctness	How often is the model right overall?
Precision	$TP/(TP+FP)$	Trust in positive predictions	When it says YES, how often is it right?
Recall	$TP/(TP+FN)$	Finding the positives	How many real positives does it find?
Specificity	$TN/(TN+FP)$	Correctly clearing negatives	How well does it avoid false alarms?
F1 Score	$2 \cdot P \cdot R / (P + R)$	Balance of P and R	Are both precision and recall decent?
ROC-AUC	area under ROC	Ranking quality	How well does it rank pos vs neg?
MAE	$\text{mean}(y - \hat{y})$	Average error (regression)	How far off is the average prediction?
RMSE	$\sqrt{\text{MSE}}$	Error, big errors penalized	How bad are the big mistakes?
R ²	$1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$	Variance explained	How much variation does it explain?

4 · FP vs FN — which error is more expensive?

FP — false alarm: model says YES, truth is NO	FN — miss: model says NO, truth is YES
spam: real email filtered out	disease: sick patient sent home
fraud: legitimate customer blocked	fraud: fraudulent payment approved
cost: annoyance, lost trust, wasted checks	cost: untreated disease, lost money, breaches
FP expensive → maximise PRECISION (raise threshold)	FN expensive → maximise RECALL (lower threshold)

5 · Golden Rules

- Never judge an imbalanced problem by accuracy alone — read precision, recall and F1.
- The threshold 0.5 is a convention, not a decision — set it from the costs of FP and FN.
- Always read the confusion matrix — a single number can hide the whole story.
- The highest metric is not automatically the best model — the business question decides.
- Judge the model only on data it has never seen.